

Project Proposal

Explainable Reinforcement Learning (XRL) for Autonomous Driving

Problem Statement

An interpretable navigation policy for intelligent driving agents trained using a world model with an explainable RL for autonomous driving in a real world setting.

Objective: To train agents on World Models (Internal WM) using reinforcement learning. Existing explainable AI (XAI): Vision Language Action Models (VLAMs) techniques does not explain the underlying “action” driving policy for agents behavior.

Q: How to provide interpretable features, policy and learning process for an agent?

Why?

- Current: Perception Architectures are based on explainable machine learning.
- Transition: LLMs (Large Language Models) → LWMs (Large World Models)
- Future: Cognitive Architectures based on explainable RL. ~ *My personal opinion.*

Existing Methods

XAI based Autonomous Vehicles

An autonomous vehicle provides a natural and intelligible explanation of its real-time decision. AVs must also explain their AI-guided decision-making process. Need to interpret the natural language based explanations for actions.

Objective

To build an agent to provide reasoning capabilities to an agent's actions.

Current Work

LINGO-1

- An open-loop driving model, on various vision and language data sources perform visual question answering (VQA) on tasks such as perception, counterfactuals, planning, reasoning and attention.
- Prompt LINGO-1 with questions regarding scene understanding and reasoning about the primary causal factors in the scene affecting the driving decision.
- It can provide a description of the driving actions and reasoning.

Need to improve driving performance through better planning and reasoning. Two factors affect driving performance

- Ability of the language model to accurately interpret scenes.
- Model to translate reasoning into effective low-level planning.

Q: How to interpret the driving policy?

LINGO-2: Driving with Natural Language

- A driving model that links vision, language, and action to explain and determine driving behavior, opening up a new dimension of control and customization for an autonomous driving experience.
- It is the first closed-loop vision-language-action driving model (VLAM) tested on public roads.

Q: How to interpret these explanations given by VLAMs?

Proposed Solution

Explainable Reinforcement Learning

Why XRL? Explainable reinforcement learning is an emerging subfield of explainable ML. •

The goal of XRL is to elucidate the decision-making process of reinforcement learning (RL) agents in sequential decision-making settings.

- What the agents will do and why.

A novel taxonomy for organizing the XRL with three high-level categories:

1. Feature Importance,
2. Learning process and Markov decision process, (LPM) and
3. Policy-level (PL)

Why? RL is combined with the generalization and representational power of deep neural networks, which is often required to achieve the desired performance on these tasks.

Proposed XRL taxonomy and its relationship to the RL problem.

Explanations may explain individual

1. actions a in terms of the input state s (Feature Importance),
2. experiences from training (Learning Process and MDP), and
3. Long-term behavior (Policy-Level)

Given a state s_t from a domain (a) , many types of explanations can be created, such as

- an object saliency map (b)
- a natural language explanation (c) and
- a decision tree policy (d)

Note that the explanations convey different information. For example, the natural language explanation (c) does not contain the counterfactual information of the decision tree policy (d) .

Progress Measures

1. First, test and run an existing VLM on an edge device with output results.
2. Second, baseline implementation of a simple world model based on RL. 3.
- Third, reproduce a baseline paper on prototype bases agent for RL. 4.
- Fourth, possibility of extending the research for explainability.
5. Lastly, discuss XRL categorization and future of RL research.